Lattices and Codes (a brief introduction)

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0 Preamble

The goal here to is to introduce irreducible root systems, and through them try to classify certain lattices. As we've seen, lattices are just about the best thing there is in number theory (maybe), since it's easy to pop out a modular form from a lattice under the right circumstances. Luckily, we'll be working in those very circumstances!

To get there, we're going to have to define what a code is, somewhat unfortunately. I'm going to avoid as much of the code talk as possible, but some of it is pretty useful, such as define the dual lattice and examining un-normalised lattices (i.e. volume not equal to 1). All the things covered in class I will attempt to summarise rather than detail.

The source of this talk is *Lattices and Codes* by Wolfgang Ebeling.

1 Lattices and Codes

1.1 Lattices

Definition 1.1. A *lattice* $\Gamma \subset \mathbb{R}^n$ is a subset such that there exists a basis $\{e_1, \ldots, e_n\}$ so that $\Gamma = \mathbb{Z}e_1 \oplus \cdots \oplus \mathbb{Z}e_n$. The *fundamental parallelotope* of Γ is

$$\mathbb{R}^n/\Gamma = P = \{\lambda_1 e_1 + \cdots + \lambda_n e_n : \lambda_i \in [0,1]\}.$$

The *volume* of a lattice is

$$\mathrm{vol}(\mathbb{R}^n/\Gamma) = \mathrm{vol}(P) = |\det((e_1, \ldots, e_n))|.$$

Under this definition, we get $\mathrm{vol}(P) = 1$ for the standard lattice $\mathbb{Z}^n \subset \mathbb{R}^n$. Given two lattices $\Gamma' \subset \Gamma \subset \mathbb{R}^n$, we view Γ' as a subgroup of Γ of finite rank, and we see

$$\mathrm{vol}(\mathbb{R}^n/\Gamma') = \mathrm{vol}(\mathbb{R}^n/\Gamma) \cdot |\Gamma : \Gamma'|.$$

Definition 1.2. Recall that $\mathbb{R}^n$ is self-dual as a vector space. Specifically, we can let $x \mapsto x \cdot -$ be the explicit isomorphism. We define the *dual lattice* of Γ, denoted Γ^*, by

$$\Gamma^* := \mathrm{Hom}(\Gamma, \mathbb{Z}) = \{x \in \mathbb{R}^n : x \cdot y \in \mathbb{Z} \text{ for all } y \in \Gamma\}.$$

We will use dot notation because it will be fairly clear when we're multiplying vectors and scalars, so no need for brackets. Note that we don't necessarily have $\Gamma \subset \Gamma^*$, but we would like to. A lattice is *integral* in this case, i.e. $x \cdot y \in \mathbb{Z}$ for all $x, y \in \Gamma$. It turns out that integral lattices are something a priori worth studying.

Proposition 1.3. The integral lattices in $\mathbb{R}^n$ are precisely the symmetric symmetric bilinear forms over $\mathbb{Z}$ for some $b : S \times S \to \mathbb{Z}$, where S is a free abelian group of rank n.

If Γ is an integral lattice with basis $\{e_i\}$, then define

$$A := ((e_i \cdot e_j)).$$

Then $\det A \in \mathbb{Z}$, as all its entries are integers. This number is actually independent of choice of basis (which should be pretty clear). In fact, we can derive from the previous discussion that

$$\sqrt{\det A} = |\Gamma^* : \Gamma|.$$

We call a lattice *unimodular* if $\Gamma^* = \Gamma$. This is also equivalent to, for an integral lattice, $\det A = 1$ or $\mathrm{vol}(P) = 1$. Hence we have been looking at unimodular lattices for a while! We need one more definition:

Definition 1.4. An integral lattice is called *even* if $x^2 = x \cdot x \in 2\mathbb{Z}$ for all $x \in \Gamma$. Equivalently, the matrix A has even elements along its diagonal.

1.2 Codes

I will restrict the discussion as much as possible to what is already relevant.

Definition 1.5. A *code of length n*, usually denoted C, is a nonempty proper subset of $\mathbb{F}_q^n$, where q is a prime power. A code is called *linear* if it is a linear subspace. Given two codewords $x, y \in C$, we can define the *(Hamming) distance* between x and y as the number of different terms, entry-wise. We might formally use $w(x - y)$ to count the number of nonzero terms; hence $d(x, y) = w(x - y)$.

Since we only really care about linear codes, we will immediately jump into that notation. We write C is an *[n,k,d]-code* if it is a dimension k subspace of $\mathbb{F}_q^n$ of minimum distance d, that is $d = \min\{d(x, y) : x, y \in C\}$. In our case, we always have $q = 2$.

We will now build up (though you don't know it yet) special types of codes analogous to lattices. We define the *dual code* to C, denoted $C^\perp$, to be

$$C^\perp := \{y \in \mathbb{F}_q^n : x \cdot y = 0 \text{ for all } x \in C\}.$$

(This isn't really the definition, but it ought to be.) A linear code C is *self-dual* if, you guessed it, $C = C^\perp$. Equivalently, C is self-dual if $\dim n$ is even, $k = n/2$, and $C \subset C^\perp$.

For a binary code, we say it is *doubly even* if $w(x) \in 4\mathbb{Z}$ for all $x \in C$, where w counts the number of nonzeroes. A doubly even code automatically satisfies $C \subset C^\perp$, which is lovely for us. One code that will be an important example is the *extended Hamming code*, denoted $\widetilde{H}$, which is a binary [8,4,4]-code which is doubly even and self-dual. More on that later.

1.3 Codes $\to$ Lattices

Henceforth all codes are linear, binary $[n, k, d]$-codes unless otherwise noted. Let ρ be the reduction of $\mathbb{Z}^n$ modulo 2. Then we may view $C \subset \rho(\mathbb{Z}^n)$, and it has index 2^{n-k}. Hence $\rho^{-1}(C) \subset \mathbb{Z}^n$ is a finite index subgroup, which makes it a lattice with $\mathrm{vol}(\mathbb{R}^n/\rho^{-1}(C)) = 2^{n-k}$.

Definition 1.6.

$$\Gamma_C := \frac{1}{\sqrt{2}}\rho^{-1}(C).$$

Why the fraction? Let's see. It's certainly a lattice in $\mathbb{R}^n$. Now take $x, y \in \Gamma_C$. Then we can write

$$x = \frac{1}{\sqrt{2}}(c + 2z), \quad y = \frac{1}{\sqrt{2}}c' + 2z'$$

for $c, c' \in C$, $z, z' \in \mathbb{Z}^n$. Now we see

$$x^2 = \frac{1}{2}(c^2 + 4cz + 4z^2) \quad \text{and} \quad x \cdot y = \frac{1}{2}(c \cdot c') \pmod{\mathbb{Z}}.$$

Then we get the following proposition.

Proposition 1.7. Let C be a (binary) linear code.

(a) $C \subset C^\perp$ if and only if Γ_C is an integral lattice.

(b) C is doubly even if and only if Γ_C is an even lattice.

(c) C is self-dual if and only if Γ_C is unimodular.

This was the entire point of mentioning codes to begin with. Now we have two dictionaries to work with in discussion lattices. We now want to talk more about lattices.

1.4 Root Lattices

We are going to classify lattices by something called irreducible root systems. It certainly sounds like a good thing to do, even if the words don't make sense yet. Consider an even (integral) lattice Γ. Then define

$$R := \{x \in \Gamma : x^2 = 2\}.$$

We call the elements *roots*.

Definition 1.8. An even lattice $\Gamma \subset \mathbb{R}^n$ is a *root lattice* if R generates Γ.

Now let's let Γ be a root lattice. Let $x, y \in R$. Then

$$(x \cdot y)^2 \leq x^2 \cdot y^2 = 4 \implies x \cdot y \in \{0, \pm 1, \pm 2\}.$$

Further, $x \cdot y = \pm 2$ if and only if $x = \pm y$, and if $x \cdot y = 1$, then $(x - y)^2 = 2$ and thus $x - y \in R$ and conversely. Now we begin to get to the point.

Theorem 1.9. Let $\Gamma \subset \mathbb{R}^n$ be a root lattice. Then Γ contains a basis $\{e_i\}$ so that

$$e_i^2 = 2 \text{ for all } i \in \{1, \dots, n\} \quad \text{and} \quad e_i \cdot -e_j \in \{0, 1\}, i \neq j, i, j \in \{1, \dots, n\}.$$

This is the goal of the section. We have the following definition which will help in the proof.

Definition 1.10. A subset $S \subset R$ is called a *fundamental system of roots* if

(a) S is a basis of Γ;

(b) Each $\beta \in R$ can be written as a linear combination

$$\beta = \sum_{\alpha \in S} k_\alpha \alpha$$

such that all k_α are non-positive or all are non-negative.

Proving that every root lattice contains a fundamental system of roots proves the theorem. We can construct these systems in general, and so let us begin to do so.

For each $\alpha \in R$, let us define the hyperplane

$$H_\alpha := \{x \in \mathbb{R}^n : x \cdot \alpha = 0\}.$$

Since R is finite, we know $\bigcup_{\alpha \in R} H_\alpha \neq \mathbb{R}^n$, so there exists some $t \in \mathbb{R}^n$ so that $t \cdot \alpha \neq 0$ for all $\alpha \in R$. We can divide R into two groups; for such a (particular) t, let

$$R_t^+ := \{\alpha \in R : t \cdot \alpha > 0\}$$

so that $R = R_t^+ \sqcup -R_t^+$. We will call R_t^+ the *positive roots*. A root $\alpha \in R_t^+$ is called *decomposable* if there exist $\beta, \gamma \in R_t^+$ so that $\alpha = \beta + \gamma$, and *indecomposable* otherwise. Let $S_t \subset R_t^+$ be the set of indecomposable positive roots.

Proposition 1.11. S_t is a fundamental system of roots for Γ. Conversely, given a fundamental system of roots S, then there exists $t \in \mathbb{R}^n$ so that $S = R_t^+$.

Proof. We will proceed in steps.

Lemma 1.12. Each element of R_t^+ is a linear combination of elements of S_t with non-negative integer coefficients.

Suppose not. Then let $\alpha \in R_t^+$ without this property. Further, we may assume we have chosen this α so that $t \cdot \alpha > 0$ is minimal (since we are working with a finite set). Then suppose α is decomposable. Let $\alpha = \beta + \gamma$. Then since we are working with positive roots,

$$t \cdot \alpha = t \cdot \beta + t \cdot \gamma \implies t \cdot \alpha > t \cdot \beta, t \cdot \gamma.$$

Therefore β, γ must be a linear combination of elements of S_t with non-negative integer coefficients. But the sum of two such combinations also has non-negative coefficients, a contradiction.

We have thus reduced to $\alpha \in S_t$ lacking this property, which is absurd.

Lemma 1.13. For $\alpha \neq \beta \in S_t$, we have $\alpha \cdot \beta \in \{0, -1\}$.

As we showed above, $\alpha \cdot \beta = \pm 2$ if and only if $\alpha = \pm\beta$ and $\alpha \cdot \beta = 1$ if and only if $\alpha - \beta \in R$. In the first case, we cannot have $\alpha, -\alpha \in S_t$, as exactly one of these may be positive. In the second case, let $\gamma = \alpha - \beta$. Then $\alpha = \beta + \gamma$ and $\beta = \alpha - \gamma$, which contradictions the indecomposability of α, β.

Lemma 1.14. The elements of S_t are linearly independent.

Suppose we had a minimal relation $\sum_{\alpha \in S_t} a_\alpha \alpha = 0$. Then splitting the k_α into positive and negative terms, we may rewrite this

$$\sum b_\beta \beta = \sum c_\gamma \gamma, \quad b_\beta, c_\gamma > 0, \{\beta\} \cap \{\gamma\} = \emptyset.$$

Let λ be this lefthand sum. Then

$$\lambda^2 = \sum_{\beta,\gamma} b_\beta c_\gamma (\beta \cdot \gamma) \geq 0$$

since $\beta \cdot \gamma \leq 0$ for each $\beta \neq \gamma \in S_t$. Therefore $\lambda = 0$. Further, all its coefficients are zero as

$$0 = t \cdot \lambda = \sum b_\beta (t \cdot \beta),$$

as each $t \cdot \beta > 0$. This will also show c_γ are all zero, so we are done. The above lemmas show that S_t is a basis for $\mathbb{R}^n$, so the fundamental system of roots is the required basis in any root lattice.

Lemma 1.15. Let $\{\gamma_1, \ldots, \gamma_n\}$ be a basis of $\mathbb{R}^n$. Then there exists an element $t \in \mathbb{R}^n$ with $t \cdot \gamma_i > 0$ for all $i \in \{1, \ldots, n\}$.

Let δ_i' be the projection of γ_i onto the subspace spanned by the other $n-1$ basis vectors, and let $\delta_i := \gamma_i - \delta_i'$. Then

$$\delta_i \cdot \gamma_j \begin{cases} > 0 & i = j \\ = 0 & i \neq j \end{cases}.$$

This is seen by appealing to the orthonormal basis associated to $\{\gamma_i\}$ by the Gram-Schmidt process. Therefore the element $t = \sum_{i=1}^n \delta_i$ fits the bill.

Now for the converse of the proposition, given a fundamental system of roots S for R, the existence of a $t \in \mathbb{R}^n$ so that $t \cdot \alpha > 0$ for all $\alpha \in S$ is guaranteed by the last lemma. Let $R^+ \subset R$ be those roots which are purely nonnegative linear combinations of elements of S. Then $R^+ \subset R_t^+$ and $-R^+ \subset -R_t^+$, so this shows $R^+ = R_t^+$. Therefore the elements of S are the indecomposable elements of R_t^+, and $S \subset S_t$. Since $|S| = |S_t| = n$, we are done. $\square$

So we come to the end, and the good question here is 'so what'? Given a fundamental system of roots $S = \{e_1, \ldots, e_n\}$, we can associate to it a graph, called a *Coxeter-Dynkin diagram*, constructed as follows: let $V = S$ be the vertex set, and draw an edge between e_i and e_j if and only if $e_i \cdot e_j = -1$. Let G be this graph. What can we say about it from our proofs so far?

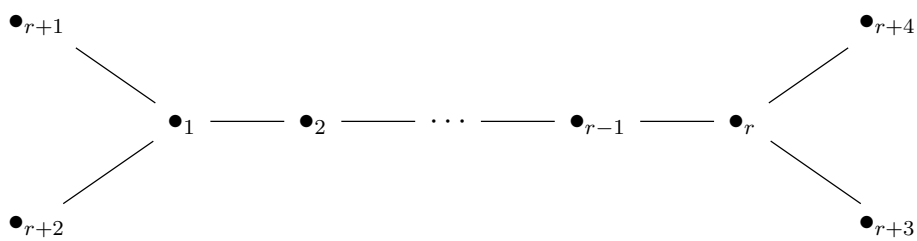
Lemma 1.16. G has no cycles.

Proof. Let $x_1 \cdots x_r$ be a minimal cycle of G, $r \geq 3$. Then we have

$$(x_1 + \cdots + x_r)^2 = \sum_{i=1}^{r} x_i^2 + \sum_{i \neq j} x_i \cdot x_j = 2r - 2r = 0.$$

But this is impossible as all roots have $x^2 = 2$. $\square$

Lemma 1.17. G does not contain a subgraph of the form (for $r \geq 1$)

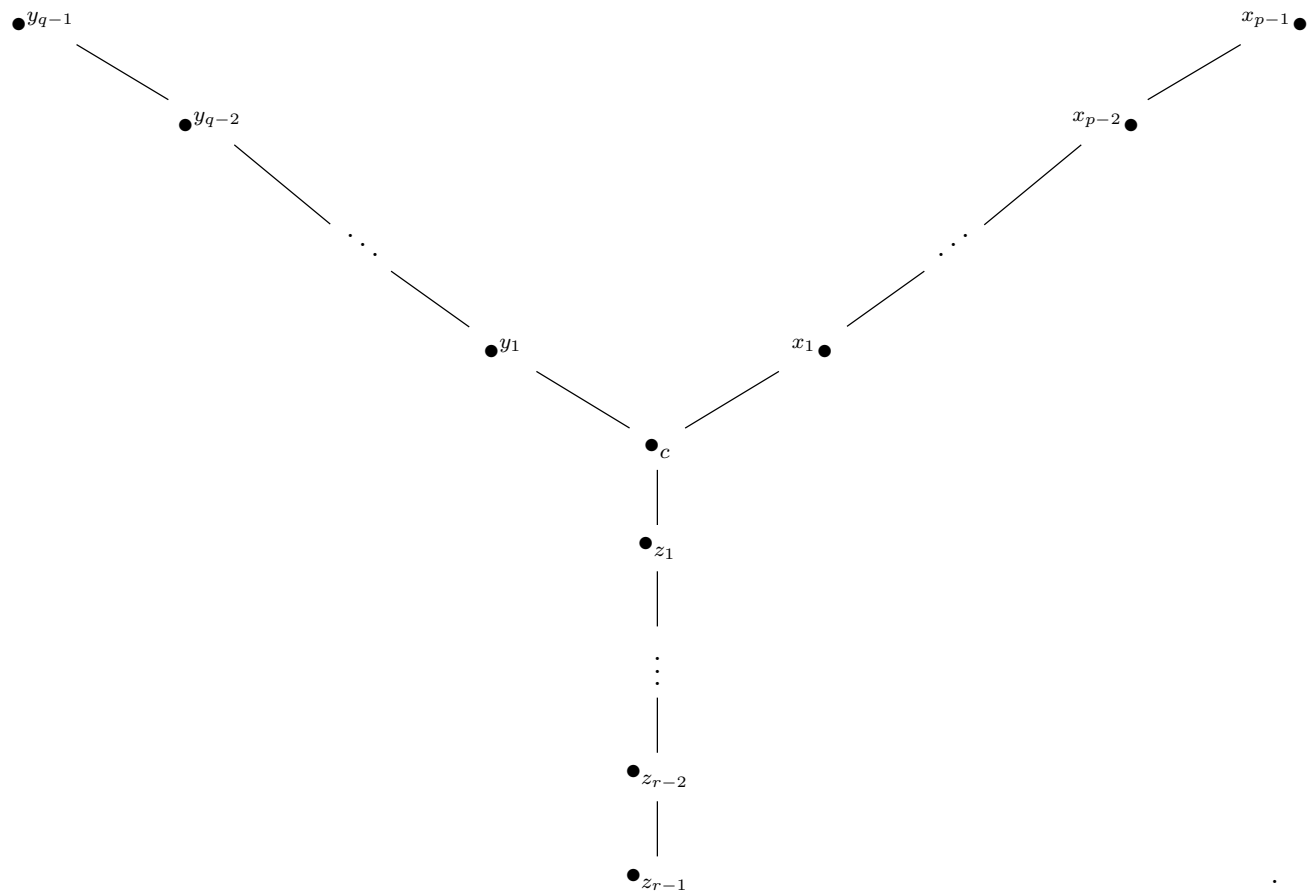


Proof. Given that graph (and throwing in x for notation), we have

$$\begin{aligned}(2x_1 + \cdots + 2x_r + x_{r+1} + \cdots + x_{r+r})^2 &= 4\sum_{i=1}^{r} x_i^2 + 4 \sum_{1 \leq i \neq j \leq r} x_i \cdot x_j + \sum_{i=r+1}^{r+4} x_i^2 \\ &\quad + 2(x_{r+4} \cdot x_r + x_{r+3} \cdot x_r + x_{r+2} \cdot x_1 + x_{r+1} \cdot x_1) \\ &= 8r + 8 - 2(r-1) - 16 = 0.\end{aligned}$$

This is also impossible. $\square$

Therefore if G is a connected graph, then it must be of the form below, for $q, p, r \geq 1$.



We can now analyse G further. Define the point

$$w := c + \frac{1}{p}\sum_{i=1}^{p-1}(p-i)x_i + \frac{1}{q}\sum_{j=1}^{q-1}(q-j)y_j + \frac{1}{r}\sum_{k=1}^{r-1}(r-k)z_k.$$

Then the collection $\{x_i, y_j, z_k, w\}$ are a $\mathbb{Q}$-basis for $\Gamma \otimes \mathbb{Q}$. Additionally, the x_i, y_j, z_k are (between each other) pairwise orthogonal, and w is orthogonal to all other basis elements. Therefore

$$w^2 = w \cdot c = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} - 1.$$

Since we must have $w^2 > 0$, we have

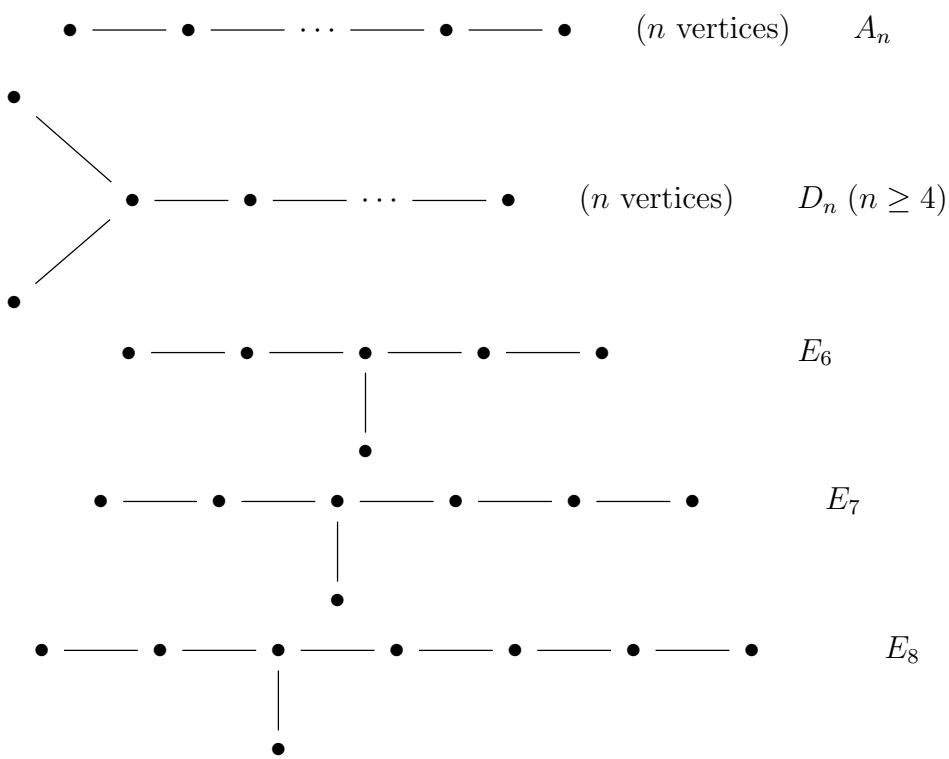
$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1.$$

This yields two general solutions and three particular ones:

$$\begin{aligned}(p,q,r) = &(1,q,r), \quad q \leq r \text{ arbitrary},\\ &(2,2,r), \quad r \text{ arbitrary},\\ &(2,3,3),\\ &(2,3,4),\\ &(2,3,5).\end{aligned}$$

A lattice Γ is called *reducible* if it can be written $\Gamma = \Gamma_1 \oplus \Gamma_2$ for two lattices $\Gamma_1 \subset \mathbb{R}^{n_1}$ and $\Gamma_2 \subset \mathbb{R}^{n_2}$ with $n_1, n_2 \geq 1$ and $n_1 + n_2 = n$. A lattice is *irreducible* if it is not reducible, and in this case its Coxeter-Dynkin diagram is connected. In general, a root lattice Γ can be decomposed as $\Gamma = \bigoplus \Gamma_i$, where Γ_i are irreducible, and the Coxeter-Dynkin diagram of Γ has as connect components the Coxeter-Dynkin diagrams of the Γ_i. Hence

Theorem 1.18. Every root lattice is the orthogonal direct sum of the irreducible root lattices with the following Coxeter-Dynkin diagrams:



These are the only theoretically possible Coxeter-Dynkin diagrams. But are there actual lattices that correspond to these? This is the question of existence, and indeed, they do exist.

To construct A_n: take the standard basis $\{e_i\}$ of $\mathbb{R}^{n+1}$. Then consider the subspace orthogonal to $e = \sum e_i$, which is isomorphic to $\mathbb{R}^n$. We may take the sublattice Γ of the standard lattice $\mathbb{Z}^{n+1} \subset \mathbb{R}^{n+1}$ contained in $e^\perp$, which is of rank n. This lattice is generated by elements of the form $e_i - e_j$ for $i \neq j$, which indeed are its roots. We have $|R| = n(n+1)$, and there is a fundamental system of roots

$$\{e_2 - e_1, e_3 - e_2, \ldots, e_{n+1} - e_i\}.$$

The Coxeter-Dynkin diagram of this basis is seen to be A_n, as (for $i, j \in \{1, \ldots, n\}$)

$$\begin{aligned}(e_{i+1} - e_i) \cdot (e_{j+1} - e_j) &= e_{i+1} \cdot e_{j+1} - e_i \cdot e_{j+1} - e_{i+1} \cdot e_j + e_i \cdot e_j \\ &= 2\delta_i(j) - \delta_i(j+1) - \delta_{i+1}(j) \\ &= \begin{cases} 2 & i = j \\ -1 & i = j \pm 1 \\ 0 & \text{else} \end{cases}.\end{aligned}$$

To construct D_n: Note that D_2 makes no sense, because to form the $\langle$ requires at least 3 vertices. Further, $D_3 \cong A_3$, so we ignore this case to make our classification unique. Therefore for $n \geq 4$, consider

$$\Gamma = \{(x_1, \ldots, x_n) \in \mathbb{Z}^n : \sum x_i \in 2\mathbb{Z}\}.$$

This is indeed a lattice, and it is generated by (for the standard basis $\{e_i\}$ of $\mathbb{R}^n$) the points $\pm e_i \pm e_j$ and $\pm e_i \mp e_j$ for $i \neq j$, of which there are $2n(n-1)$. Then we have a fundamental system of roots

$$\{e_2 - e_1, \ldots, e_n - e_{n-1}, e_1 + e_2\}.$$

Then we have A_{n-1} embedded as a subgraph corresponding to the obvious basis elements. We need only check $e_1 + e_2$ against an arbitrary element: for $i \in \{1, \ldots, n-1\}$,

$$\begin{aligned}(e_1 + e_2) \cdot (e_{i+1} - e_i) &= e_1 \cdot e_{i+1} - e_1 \cdot e_i + e_2 \cdot e_{i+1} - e_2 \cdot e_i \\ &= \delta_1(i+1) - \delta_1(i) + \delta_2(i+1) - \delta_2(i) \\ &= \begin{cases} -1 & i = 2 \\ 0 & \text{else} \end{cases}.\end{aligned}$$

Therefore we attach the vertex $e_1 + e_2$ to the second point in A_{n-1}, which forms D_n.

It turns out that the extended Hamming code $\widetilde{H}$ described above generates E_8. It is now time to write it down: we have

$$\begin{aligned}\widetilde{H} := \operatorname{span}\{&(0,1,1,0,1,0,0,1), (0,0,1,1,0,1,0,1), (0,0,0,1,1,0,1,1), (1,0,0,0,1,1,0,1) \\ &(0,1,0,0,0,1,1,1), (1,0,1,0,0,0,1,1), (1,1,0,1,0,0,0)\} \subset \mathbb{F}_2^7.\end{aligned}$$

Recall that we define $\Gamma_{\widetilde{H}}$ by $\frac{1}{\sqrt{2}}$ times the pullback of the code to $\mathbb{R}^8$, which we have already shown is an even unimodular lattice (viz. the most interesting kind of lattice). Let $f_1, \ldots, f_7$ be defined by the above vectors scaled by $\frac{1}{\sqrt{2}}$. Then we have $f_i^2 = 2$ for each vector, and $f_i \cdot f_j = 1$ for $i \neq j$. From here, we can (begin to) construct a fundamental root system for $\Gamma_{\widetilde{H}}$: let $e_1 = f_1$ and for $i > 1$, let $e_i = f_i - f_{i-1}$. Together, these elements form a subgraph isomorphic to A_7 of the diagram of $\Gamma_{\widetilde{H}}$. However, we need one more basis element to finish off the construction. We take $e_8 := \frac{1}{\sqrt{2}}(-1, -1, 0, 0, 1, 0, -1, 0)$, which satisfies

$$e_8^2 = 2, \quad e_8 \cdot f_i = \begin{cases} 0 & i = 1,2,3,4 \\ -1 & i = 5,6,7 \end{cases}.$$

Hence $e_8 \cdot e_i = -1$ if and only if $i = 5$ and $e_8 \cdot e_i = 0$ otherwise. This yields E_8. A slight modification of this construction will also yield E_7 and E_6.

At this point it is worth noting that we are missing a few letters in the above, obviously B and C, but also F and G, though you wouldn't know it. The above process is precisely how one goes about classifying simple Lie algebras, of which there are four infinite families and five exceptional ones (and maybe I will write these down at this point). In the general, case, the graphs considered are directed (as the inner product defined on the roots is not symmetric), and we have situations where $\alpha_i \cdot \alpha_j \in \{\pm 2, \pm 1, 0\}$ are all possible. The reader should go look this up if it's interesting.

1.5 The Weyl Group

Let Γ be a root lattice in $\mathbb{R}^n$, and let $\alpha \in R$ be a root. Consider the linear transformation

$$s_\alpha : \mathbb{R}^n \to \mathbb{R}^n, \qquad x \mapsto x - 2(x \cdot \alpha)\alpha$$

the reflection of $\mathbb{R}^n$ about the hyperplane $\alpha^\perp$. This is an invertible linear transformation, so lives in $\mathrm{GL}_n(\mathbb{R})$. Let $W(\Gamma) \subset \mathrm{GL}_n(\mathbb{R})$ be the subgroup generated by s_α for all $\alpha \in R$.

Lemma 1.19. Let $\Gamma \subset \mathbb{R}^n$ be an irreducible root lattice. Then $\mathbb{R}^n$ is irreducible as a $W(\Gamma)$-module.

Proof. Suppose $U \subset \mathbb{R}^n$ is a $W(\Gamma)$-submodule. Then since $\mathbb{R}[W(\Gamma)]$ is a semisimple algebra, $U^\perp$ is a $W(\Gamma)$-submodule as well. Let $\alpha \in \Gamma$ be a root, and suppose that $\alpha \notin U$. Let $u \in U$ be any element. Since $s_\alpha(U) = U$, we have $u - (u \cdot \alpha)\alpha \in U$, whence $u \cdot \alpha = 0$ else we find $\alpha \in U$. Therefore $\alpha \in U^\perp$. Therefore every root lies in U or $U^\perp$. Since R spans Γ, $\Gamma = (U \cap \Gamma) \oplus (U^\perp \cap \Gamma)$. But Γ was assumed to be irreducible, so $U \cap \Gamma = \emptyset$ or $U \cap \Gamma = \Gamma$, which implies that $U = 0$ or $\mathbb{R}^n$. □

Lemma 1.20. Let $\Gamma \subset \mathbb{R}^n$ be a root lattice. Then $W(\Gamma)$ acts transitively on the set of roots R.

Proof. Let $\alpha', \beta \in R$ be two roots. Then the set $\{w(\alpha') : w \in W(\Gamma)\}$ spans $\mathbb{R}^n$, as it is a nonzero $W(\Gamma)$-submodule of $\mathbb{R}^n$. Therefore choose some $\alpha := w(\alpha')$ not orthogonal to β. If $\alpha = \beta$, then we are done as we have found an element of the Weyl group transformation any root into another.

Hence suppose $\alpha \neq \beta$, so $\alpha \cdot \beta = \pm 1$. Without loss of generality, we may replace β by $s_\beta(\beta)$ so that $\alpha \cdot \beta = 1$. Then

$$s_\alpha s_\beta s_\alpha(\beta) = s_\alpha s_\beta(\beta - (\beta \cdot \alpha)\alpha) = s_\alpha(\beta - \alpha - \beta) = \alpha.$$

This completes the proof. □

This may be a terrible aside (as it is not in Ebeling's book), but the following discussion is very interesting, which I presented in the Fall for the Algebraic Groups seminar. The extra material comes from Humphreys' *Linear Algebraic Groups*.

Viewing a root lattice $\Gamma \subset \mathbb{R}^n$ with roots R, we can consider $\operatorname{Aut} R := \{\sigma \in \mathrm{GL}_n(\mathbb{R}) : \sigma(R) = R\}$, those automorphisms of $\mathbb{R}^n$ that preserve the roots. It is clear that $W(\Gamma) \subset \operatorname{Aut} R$. Further, we can define the automorphism group of the Coxeter-Dynkin diagram G of Γ, denoted $\operatorname{Aut} G$, by permutations of the vertices that preserve the edge relations. Further, we can view an automorphism of G as an automorphism of R by pulling back the permutation of the vertices indexed by a fundamental system of roots S to a permutation of S as a generating set of R. This gives us a sequence of groups

$$1 \to W(\Gamma) \to \operatorname{Aut} R \to \operatorname{Aut} G \to 1.$$

Proposition 1.21. (Humphreys 12.2) The above sequence is exact.

Proof. We need a few lemmas to prove this.

Lemma 1.22. (Humphreys 9.2) $W(\Gamma)$ is a normal subgroup of $\operatorname{Aut} R$.

It suffices to show that the set of generators $\{s_\alpha\}$ is fixed under conjugation by elements of $\operatorname{Aut} R$. Let $\sigma \in \operatorname{Aut} R$. Then note that, for a root β,

$$\sigma s_\alpha \sigma^{-1}(\sigma(\beta)) \in R$$

since $s_\alpha(\beta) \in R$. But

$$\begin{aligned}\sigma s_\alpha \sigma^{-1}(\sigma(\beta)) &= \sigma(\beta - (\beta \cdot \alpha)\alpha) = \sigma(\beta) - (\beta \cdot \alpha)\sigma(\alpha) \\ &= \sigma(\beta) - (\sigma(\beta) \cdot \sigma(\alpha))\sigma(\alpha),\end{aligned}$$

where we use the fact that $\alpha \cdot \beta = \sigma(\alpha) \cdot \sigma(\beta)$, which I believe is reasonably clear, though caused much arguing when first presented. Since $\sigma(\beta)$ runs over R as β runs over R, we have shown that $\sigma s_\alpha s^{-1}$ fixes R. Moreover, it sends $\sigma(\alpha)$ to $-\sigma(\alpha)$ and fixes the hyperplane orthogonal to $\sigma(\alpha)$, so it is equal to $s_{\sigma(\alpha)}$.

Recall that the choice S_t of a fundamental system of roots depended on the choice of $t \in \mathbb{R}^n$, but we proved that any choice would do. Of course by a simple counting argument for a given system S there are many many $t \in \mathbb{R}^n$ such that $S_t = S$. Hence we can consider the set $\mathcal{S}$ of fundamental systems of roots of R (of a lattice Γ). There is an obvious action of $W(\Gamma)$ on $\mathcal{S}$ where

$$w : \{e_1, \ldots, e_n\} \mapsto \{w(e_1), \ldots, w(e_n)\},$$

but it is not a priori clear that what we get is still a fundamental system of roots. Fix a system of roots $S = \{e_i\}$. Let $w \in W(\Gamma)$. Then since $W(\Gamma)$ acts transitively on the set of roots, for each $\alpha \in R$ there exists $\beta \in R$ so that $w(\beta) = \alpha$. Thus if we have

$$\sum_{i=1}^n k_i w(e_i) = \alpha = w(\beta) \implies w\left(\sum_{i=1}^n k_i e_i\right) = w(\beta) \implies k_i \geq 0 \text{ or } k_i \leq 0,$$

so this (along with the other discussion above) shows that this action makes sense at all.

Lemma 1.23. (Humphreys 10.3) The above action of $W(\Gamma)$ on $\mathcal{S}$ is transitive. Additionally, the action of $W(\Gamma)$ on a particular fundamental system of roots S is simply transitive, that is for any $e_i, e_j \in S$, there is a unique $w \in W(\Gamma)$ so that $w(e_i) = e_j$.

This is a fairly nontrivial proof, and since I am likely running over time already, I will omit it.

To complete the proof, we will show that $\operatorname{Aut} R$ is a semidirect product of $\operatorname{Aut} G$ and $W(\Gamma)$. For a fixed basis S, define the subgroup $H := \{\sigma \in \operatorname{Aut} R : \sigma(S) = S\} \subset \operatorname{Aut} R$. This subgroup is isomorphic to $\operatorname{Aut} G$, as it is precisely those automorphisms preserving the vertices and edge relations of G. To show that $\operatorname{Aut} R = W(\Gamma) \rtimes H$, it is necessary and sufficient to show that $W(\Gamma) \triangleleft \operatorname{Aut} R$(which we have done), $H \cap W(\Gamma) = \{1\}$, and $W(\Gamma) \cdot H = \operatorname{Aut} R$.

Suppose that $\sigma \in H \cap W(\Gamma)$. By the second lemma, we have $\sigma = 1$ by simple transitivity. Now for any $\tau \in \operatorname{Aut} R$, we have shown that $\tau(S)$ is still a fundamental system of roots for R. Since $W(\Gamma)$ acts transitively on $\mathcal{S}$, there exists w so that $w(\tau(S)) = S$. Therefore $w\tau \in H$, so $\tau \in W(\Gamma) \cdot H$. This completes the proof. $\square$

This allows us to easily calculate the Weyl groups for A_n, since these graphs have very simple automorphism groups. Indeed, we have $\operatorname{Aut} G = \mathbb{Z}/2\mathbb{Z}$ for each of these graphs. Recall that the roots for A_n (in the construction above) are $R = \{e_i - e_j : 1 \leq i \neq j \leq n+1\}$. We may permute these roots by S_{n+1}, giving the embedding $S_{n+1} \subset \operatorname{Aut} R$. Further, there is the map $e_i - e_j \mapsto e_{n+1-i} - e_{n+1-j}$, which is not given by any permutation, which corresponds to the 'flip' of the Coxeter-Dynkin diagram. This gives $\operatorname{Aut} R \cong S_{n-1} \rtimes \mathbb{Z}/2\mathbb{Z}$, and $W \cong S_{n+1}$.

There are a few more interesting things to say about root systems in general. The index of $\Gamma \subset \Gamma^*$ and something called the 'highest root' of a root lattice would be fun to discuss, but fearing lack of time, I'm not going to bother writing anything at this time. Thus ends our discussion of root lattices per se.

2 Even Unimodular Lattices

2.1 Motivation

As stated a couple times, these are the most interesting kinds of lattices. Recall that for a general lattice $\Gamma \subset \mathbb{R}^n$, we associate the *theta function*

$$\vartheta_\Gamma(\tau) := \sum_{x \in \Gamma} q^{\frac{1}{2}x^2}$$

defined on the upper half plane $\mathbb{H}$. If Γ is an even integral lattice, then $\frac{1}{2}x^2 \in \mathbb{N}$ for all $x \in \Gamma$. Therefore we may write

$$\vartheta_\Gamma(\tau) = \sum_{r=0}^{\infty} a_r q^r,$$

where $a_r = \#\{x \in \Gamma : x^2 = 2r\}$. We've further proved in class that for an even unimodular lattice $\Gamma \subset \mathbb{R}^n$, we have $n \equiv 0 \pmod 8$ and ϑ_Γ is a modular form of weight $n/2$, in the sense that

$$\vartheta_\Gamma\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{n/2}\vartheta_\Gamma(\tau).$$

Using all the notation from class, let M_k be the space of weight k modular forms (with trivial character and with respect to the full modular group). Then $M := \oplus_{k=0}^{\infty} M_k$ is a graded $\mathbb{C}$-algebra. We proved the following proposition in class:

Proposition 2.1.

$$M \cong \mathbb{C}[E_4, E_6],$$

as a $\mathbb{C}$-algebra, where

$$E_4(\tau) = 1 + 240\sum_{r=1}^{\infty} \sigma_3(r)q^r, \quad E_6(\tau) = 1 - 504\sum_{r=1}^{\infty} \sigma_5(r)q^r$$

are the usual Eisenstein series.

This gives the following corollary.

Corollary 2.2. Let Γ be an even unimodular lattice in $\mathbb{R}^8$. Then Γ is isomorphic to E_8.

Proof. Let ϑ_Γ be the theta function of Γ. This is a modular form of weight 4, and we know that $M_4 = \mathbb{C} \cdot E_4$ as a $\mathbb{C}$-vector space. Since we know that ϑ_Γ has constant term 1, we know that $\vartheta_\Gamma = E_4$ on the nose. Therefore $a_1 = \#\{x \in \Gamma : x^2 = 2\} = 240$, so Γ has 240 roots. These roots generate a root sublattice of rank at most 8. However, we calculated the number of roots in A_8 and D_8 to be 72 and 102, respectively, and it can be shown E_6 and E_7 have 72 and 126 roots, respectively. Hence these roots must generate E_8, and thus $\Gamma = E_8$. $\square$

2.2 Theta Functions with Spherical Coefficients

The goal now will be to classify even unimodular lattices of rank ≤ 24 via their fundamental systems of roots by studying a special class of theta functions.

Definition 2.3. Let $P \in \mathbb{C}[x_1, \ldots, x_n]$ be a homogeneous complex polynomial of degree r. We say that P is *spherical of degree* r if $\Delta P = 0$, where

$$\Delta = \sum_{i=1}^{n} \frac{\partial}{\partial x_i^2}$$

is the Laplace operator.

We have a convenient characterisation of these polynomials.

Theorem 2.4. $P \in \mathbb{C}[x_1, \ldots, x_n]$ is spherical of degree r if and only if P is a linear combination of functions of the form $(\xi \cdot x)^r$, where $\xi \in \mathbb{C}^n$, $\xi^2 = 0$ if $r \geq 2$.

The proof of this theorem is not particularly enlightening for the present discussion. Now we can construct functions on the upper half plane based around spherical polynomials.

Definition 2.5. Let $\Gamma \subset \mathbb{R}^n$ be a lattice, $z \in \mathbb{R}^n$ a point, and P a spherical polynomial of degree r. Then we define

$$\vartheta_{z+\Gamma,P}(\tau) := \sum_{x \in z+\Gamma} P(x) e^{\pi i \tau x^2} = \sum_{x \in \Gamma} P(x+z) e^{\pi i \tau (x+z)^2}.$$

These functions are holomorphic in $\mathbb{H}$ and satisfy $\vartheta_{z+\Gamma,P} = \vartheta_{z'+\Gamma,P}$ if $z - z' \in \Gamma$. Ideally, these functions are modular forms, so we can apply the corpus of knowledge we have built up. Hence, we examine how they transform with respect to the modular group.

Proposition 2.6. We have the identity

$$\vartheta_{z+\Gamma,P}\left(-\frac{1}{\tau}\right) = \left(\sqrt{\frac{\tau}{i}}\right)^{n+2r} i^{-r} \frac{1}{\mathrm{vol}(\mathbb{R}^n/\Gamma)} \sum_{y \in \Gamma^*} P(y) e^{2\pi i y \cdot z} e^{\pi i \tau y^2}.$$

This follows via Poisson summation and the relative niceness of spherical polynomials. Now we make the assumption that Γ is an even integral lattice, n is even, and we define $k := n/2+r$. In this case, $\Gamma \subset \Gamma^*$, and for $y, y' \in \Gamma^*$, we have $y - y' \in \Gamma \implies y \cdot x - y' \cdot x \in \mathbb{Z}$ and $y^2 - y'^2 \in 2\mathbb{Z}$. Therefore we have the (much more useful) formulae, if we let $\rho \in \Gamma^*$:

$$\begin{aligned}
\vartheta_{\rho+\Gamma,P}(\tau+1) &= e^{\pi i \rho^2} \vartheta_{\rho+\Gamma,P}(\tau), \\
\vartheta_{\rho+\Gamma,P}\left(-\frac{1}{\tau}\right) &= \frac{1}{\operatorname{vol}(\mathbb{R}^n/\Gamma)} \left(\frac{\tau}{i}\right)^k i^{-r} \sum_{\sigma \in \Gamma^*/\Gamma} e^{2\pi i \sigma \cdot \rho} \vartheta_{\sigma+\Gamma,P}(\tau).
\end{aligned}$$

This is not quite what we would like. Therefore we find a subgroup of the modular group under which these theta functions transform more favourably. Something that I did not mention earlier is that, given a binary linear code C, we have $2\Gamma^* \subset \Gamma$. We want to do something similar in the general case.

Definition 2.7. The *level* of Γ, denoted N, is the least $N \in \mathbb{N}$ so that $N\mu^2 \in 2\mathbb{Z}$ for all $\mu \in \Gamma^*$.

Lemma 2.8. If N is the level of Γ,

$$N\Gamma^* \subset \Gamma.$$

We define the subgroups

$$\begin{aligned}
\Gamma_0(N) &:= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : N \mid c \right\}, \\
\Gamma(N) &:= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \mod N \right\},
\end{aligned}$$

where N is the *level* of Γ. Since we alluded to this in class, I will skip the foreplay and get right to the point.

Theorem 2.9. For $\Gamma(N)$, $\vartheta_{\rho+\Gamma,P}$ is a modular form of weight k. In particular, if $|_k$ is the usual notation,

$$\begin{aligned}
\vartheta_{\rho+\Gamma,P}|_k A &= \vartheta_{\rho+\Gamma,P} && A \in \Gamma(N), \\
\vartheta_{\Gamma,P}|_k A &= \left(\frac{\Delta}{d}\right) \vartheta_{\Gamma,P} && A \in \Gamma_0(N),
\end{aligned}$$

where $(-)$ is the Legendre symbol and $\Delta = (-1)^{n/2}|\Gamma^* : \Gamma| = (-1)^{n/2} \operatorname{vol}(\mathbb{R}^n/\Gamma)^2$.

2.3 Root Systems in Even Unimodular Lattices

That last section was perhaps somewhat rushed, but the point was to make it this far.

Corollary 2.10. Let $\Gamma \subset \mathbb{R}^n$ be an even unimodular lattice, and let P be as spherical polynomial in n variables of degree r. Then $\vartheta_{\Gamma,P}$ is a modular form of weight $n/2 + r$, and a cusp form if $r > 0$.

Proof. The first assertion follows because, for an even unimodular lattice, $N = 1$ and $\Gamma^* = \Gamma$, and (as we have proved) $n \equiv 0 \pmod 8$. Now we know

$$\vartheta_{\Gamma,P}(\tau) = \sum_{s=0}^{\infty} c_s q^s,$$

where $q = e^{2\pi i\tau}$, and

$$c_s := \sum_{\substack{x\in\Gamma \\ x^2=2s}} P(x).$$

If $r > 0$, then $c_0 = P(0) = 0$, so $\vartheta_{\Gamma,P}$ is a cusp form. □

We want to use this to classify even unimodular matrices of dimension ≤ 24, since modular forms of weight ≤ 12 are pretty much the only thing we have a terrific handle on. Let Γ be such a lattice and let R be its set of roots. Let

$$f(x) := (x \cdot y)^2 - \frac{x^2 \cdot y^2}{n}$$

for some fixed $y \in \mathbb{R}^n$. Then f is a spherical polynomial of degree 2, and hence $\vartheta_{\Gamma,f}$ is a cusp form of weight $n/2 + 2$. So for $n = 8, 16, 24$, we have a cusp form of weight 6, 10, and 14 respectively. But recall that for forms of weight 6 and 10, $M_k = \mathbb{C} \cdot E_k$, and this turns out to be true as well for $k = 14$. Thus if we write

$$\vartheta_{\Gamma,f} = \sum_{r=0}^{\infty} c_r q^r,$$

we have $c_r = 0$ for all r. Recalling from a few lines ago what these coefficients mean, we have

Proposition 2.11. For an even unimodular lattice $\Gamma \subset \mathbb{R}^n$ for $n = 8, 16, 24$, we have for a fixed $y \in \mathbb{R}^n$

$$\sum_{\substack{x\in\Gamma \\ x^2=2r}} (x \cdot y)^2 - \left(\sum_{\substack{x\in\Gamma \\ x^2=2r}} x^2 \right) \frac{y^2}{n} = 0.$$

In particular,

$$\sum_{x\in R} (x \cdot y)^2 = \frac{y^2}{n} \cdot 2|R|.$$

Corollary 2.12. Either $R = \emptyset$ or R spans $\mathbb{R}^n$.

Proof. If R does not span $\mathbb{R}^n$, then let $0 \neq y \in \mathbb{R}^n$ which is orthogonal to all the elements of R. Then the above formula proves $|R| = 0$. □

Corollary 2.13. Let $\Gamma \subset \mathbb{R}^n$ be an even unimodular lattice, and let $n = 8, 16, 24$. Then all irreducible components of the root lattice spanned by R have the same *Coxeter number* h, where $h = \frac{1}{n}|R|$.

Proof. This proof comes from the above proposition and the following one:

Proposition 2.14. Let $\Gamma \subset \mathbb{R}^n$ be an irreducible root lattice. Then for a fixed $y \in \mathbb{R}^n$, we have

$$\sum_{x \in R} (x \cdot y)^2 = 2 \cdot h \cdot y^2,$$

We will take this proposition on faith. □

Now what does this mean? We have already shown that for $n = 8$, any even unimodular matrix is isomorphic to E_8. Take $n = 16$, and let $\Gamma \subset \mathbb{R}^{16}$ be an even unimodular lattice. Then θ_Γ is a modular form of weight 8, and we have shown that $\vartheta_\Gamma = E_4^2$. Therefore $|R| = 480$, as we can directly calculate the coefficient of q of E_4^2 given we know

$$E_4(\tau) = 1 + 240q + \cdots$$

From the above, we see that R spans $\mathbb{R}^{16}$ and each irreducible component of the root lattice spanned by R, which we denote $R_\mathbb{Z}$ have Coxeter number 30. Since we have exhausted the possibilities for the irreducible components, we can see that

$$R_\mathbb{Z} = E_8 + E_8 \text{ or } D_{16}.$$

Proposition 2.15. Let $\Gamma \subset \mathbb{R}^{24}$ be an even unimodular lattice. Then $R_\mathbb{Z}$ is one of the following 24 lattices:

$$\begin{aligned}
&0, 24A_1, 12A_2, 8A_3, 4A_6, 3A_8, 2A_{12}, A_{24}, \\
&6D_4, 4D_6, 2D_{12}, D_{24}, 4E_6, 3E_8, \\
&4A_5 + D_4, 2A_7 + 2D_5, 2A_9 + D_6, A_{15} + D_9, \\
&E_8 + D_{16}, 2E_7 + D_{10}, E_7 + A_{17}, E_6 + D_7 + A_{11}.
\end{aligned}$$

Proof. These are the only possibilities for root systems R that satisfy the following properties:

(a) R spans $\mathbb{R}^{24}$;

(b) All of the irreducible components of $R_\mathbb{Z}$ have the same Coxeter number.

If we let

$$R_\mathbb{Z} = \sum_{i=1}^{24} \alpha_i A_i + \sum_{j=1}^{24} \delta_j D_j + \sum_{k=6}^{8} \eta_k E_k,$$

then condition (a) means that

$$24 = \sum_{i=1}^{24} i \cdot \alpha_i + \sum_{j=1}^{24} j \cdot \delta_j + \sum_{k=6}^{8} k \cdot \eta_k.$$

Condition (b) gives the following quantities, all equal to 24, each of which corresponds to a set of solutions:

$$\begin{aligned}
&i \cdot \alpha_i, j \cdot \delta_j, k \cdot \eta_k, (2j-3)\alpha_{2j-3} + j \cdot \delta_j \\
&11\alpha_{11} + 7\delta_7 + 6\eta_6, 17\alpha_{17} + 10\delta_{10} + 7\eta_7, 16\delta_{16} + 8\eta_8
\end{aligned}$$

And that is all! □

We will state the last two steps of the classification, as they involve some things about codes that's not worth talking about. Recall that for a code C, we associate a lattice Γ_C as $\frac{1}{\sqrt{2}}\rho^{-1}(C)$.

Theorem 2.16. The correspondence $C \mapsto \Gamma_C$ induces a one-to-one correspondence between equivalence classes of doubly even codes C of $\mathbb{F}_2^n$ and isomorphism classes of even lattices in $\mathbb{R}^n$ containing a root lattice of type nA_1. Self-dual codes correspond to unimodular lattices.

Theorem 2.17. Up to isomorphism, there exist precisely 24 even unimodular lattices in $\mathbb{R}^{24}$. Each lattice is uniquely determined by its root sublattice. The possible root sublattices are listed above.